Revision-Note Review Worksheet for Expert Rater

This booklet contains one student revision note to review. It is a take-home revision note prepared by a teacher for a JSS1 Mathematics class on Three dimensional figures: properties of cylinders and spheres, and volume. Judge how well it covers the intended objectives and curriculum, follows the teacher's lesson plan, stays on topic and is conceptually correct, and is clear and useful for the students.

Lesson and revision-note context

Subject	Mathematics
Grade	JSS1
Term	Third Term
Week	5
Topic	Three dimensional shapes II
Subtopic	Three dimensional figures: properties of cylinders and spheres, and volume
Theme	—
Curriculum objectives	• identify properties of cylinders and spheres • find the volume of a cube, a cuboid and a prism
Previous knowledge	• properties of cubes and cuboid • properties of pyramids and cones
Lesson sequence	• identify properties of cylinders and spheres — the teacher displays shapes of cylinders and spheres; the student identify the properties; the teacher draws the shapes on the board and adds to the properties already mentioned by the students. Set for the students: 1. How many circular faces does a solid cylinder have? 2. A cylinder has how many faces. • find the volume of a cube, a cuboid and a prism — the teacher explains how to derive the formula for volume of cube and cuboid, and works the volume of a prism as area of base times height; the student listen and ask questions where they do not understand; the student copy worked examples into their notes
Instructional materials	• shapes of cylinders and spheres.
Evaluation	• Calculate the volume of a cuboid 8cm long, 4cm wide and 3cm high. • Calculate the volume of cuboid with dimensions 5m by 2m by 3m. • What is the volume of a cube of 4cm × 4cm × 4cm. • A rectangular tank hold 6m long by 2m wide holds $36m^3$ of water. How deep is the water in the tank.

Lesson objectives

1.	identify properties of cylinders and spheres
2.	find the volume of a cube, a cuboid and a prism

Read the student revision note on the following pages, then complete the rating sheet at the end of this booklet.

STUDENT REVISION NOTE

Topic: Three dimensional figures: properties of cylinders and spheres, and volume

Class: JSS1 **Subject:** Mathematics **Week:** 5

Section 1: Cylinders and Spheres: Their Properties

In this section you will learn to identify the properties of two solid shapes: the cylinder and the sphere. You already know the cube, cuboid, pyramid and cone; now we look at shapes whose faces curve.

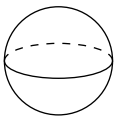
A **cylinder** looks like a tin can or a drum. Look at the cylinder in fig-01: it has a circular top, a circular base, and a curved side joining them. From the figure you can read off its properties:

1. It has **two circular faces** (the top and the base) and **one curved face**.
2. It has **no vertex** (no corner where edges meet).
3. It has **two curved edges** (the circles where the curved face meets the top face and the base).
4. Altogether it has **three faces**.

A **sphere** looks like a ball. Look at the sphere in fig-02. Its properties are:

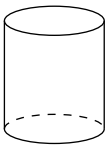
1. It is **symmetrical from all directions** — turn it any way and it looks the same.
2. It has **no edge** and **no vertex**.
3. It has **one curved surface**.

Visual



A sphere drawn as a circle with an ellipse across its middle to show the curved surface.

Visual



A cylinder drawn upright: an ellipse for the top face, two vertical sides falling from its ends, and a curve closing the base.

Let us compare a cylinder and a sphere side by side.

Example. Count the faces, edges and vertices of a cylinder.

In fig-01 you can see the two circular ends and the curved side between them. So the cylinder has:

- Faces: two circular faces plus one curved face = **3 faces**
- Edges: two curved edges (where the curved face meets each circular face)
- Vertices: none

Example. Count the faces, edges and vertices of a sphere.

In fig-02 the sphere has only its curved surface. So:

- Faces: **1 curved face**
- Edges: none
- Vertices: none

Notice the difference: the cylinder has two curved edges and no vertex, while the sphere has no edge at all.

Practice

1. How many circular faces does a solid cylinder have?
2. A cylinder has how many faces altogether?

3. How many vertices does a sphere have?
4. Name two properties that tell a cylinder apart from a sphere.

Section 2: Volume of Cubes, Cuboids and Prisms

The volume of a solid tells you how much space it takes up. You already know the units — when you measure in centimetres, volume is in cubic centimetres (cm^3), and when you measure in metres, it is in cubic metres (m^3).

For a cuboid, the volume is found by multiplying the area of its base by its height:

Volume of cuboid = area of base × height
= length × breadth × height
= lbh

Remember that length, breadth and height are all measured in the same unit, and you multiply them together.

A cube is a special cuboid where every edge is the same length, l. So all three dimensions equal l, and

Volume of cube = $l \times l \times l = l^3$

You can read l^3 as "l cubed".

The same idea — volume equals area of base × height — also works for a prism. A prism has two identical parallel faces joined by a constant cross-section. So for any prism:

Volume of prism = area of base × height

You first find the area of the base face, then multiply it by the height (the distance the prism runs).

Example 1. Find the volume of a cuboid of length 3 cm, breadth 2 cm and height 8 cm.

Volume of cuboid = L × b × h

length = 3 cm, breadth = 2 cm, height = 8 cm

Volume = 3 cm × 2 cm × 8 cm
= 48 cm^3

So the cuboid has volume 48 cm^3.

Example 2. A cuboid has volume 24 cm^3. Its base is 6 cm by 2 cm. Find its height.

Volume of cuboid = L × b × h

length = 6 cm, breadth = 2 cm, height =?

Volume = 24 cm^3

$24\ cm^3 = 6\ cm \times 2\ cm \times h$
$24\ cm^3 = 12\ cm^2 \times h$

Divide both sides by 12 cm^2:

$24\ cm^3 \div 12\ cm^2 = h$
$2\ cm = h$

So the height is 2 cm.

Example 3. Find the volume of a cube of side 3 cm.

Volume of cube = l^3

$V = (3\ cm)^3$
$V = 3\ cm \times 3\ cm \times 3\ cm$
$V = 27\ cm^3$

So the cube has volume 27 cm^3.

Example 4. Find the volume of the triangular prism. Its triangular end has a base of 5 m and a height of 3 m, and the prism runs back 10 m.

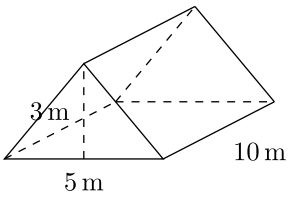
First find the area of the triangular base:

Area of triangle = ½ × b × h
= ½ × 5 × 3
= 7.5 m²

Then volume = area of base × height
= 7.5 m² × 10 m
= 75 m³

So the prism has volume 75 m³.

Visual



A triangular prism lying on its rectangular face. The triangular end has a base of 5 m and a height of 3 m, and the prism runs back 10 m.

Practice

1. Calculate the volume of a cuboid 8 cm long, 4 cm wide and 3 cm high.
2. Calculate the volume of a cuboid with dimensions 5 m by 2 m by 3 m.
3. What is the volume of a cube of side 4 cm?
4. A rectangular tank is 6 m long and 2 m wide, and it holds 36 m³ of water. How deep is the water in the tank?

Check your work

Attempt every question before you read this page.

Cylinders and Spheres: Their Properties

1. Two.

   A solid cylinder has a circular top and a circular base. Each is a circular face, so there are 2 circular faces.
2. Three faces (two circular faces and one curved face).

   The cylinder has two circular faces (top and base) plus one curved face joining them. So altogether it has 2 + 1 = 3 faces.
3. None — a sphere has no vertex and no edge.

   A sphere is only a curved surface, so it has no corners and no edges.
4. A cylinder has two circular faces and two curved edges; a sphere has one curved surface and no edges.

   The cylinder has faces that are circles and two curved edges, while the sphere has only a curved surface with no edges and no vertices.

Volume of Cubes, Cuboids and Prisms

1. 96 cm³

   Volume = L × b × h = 8 cm × 4 cm × 3 cm = 96 cm³.
2. 30 m³

   Volume = L × b × h = 5 m × 2 m × 3 m = 30 m³.
3. 64 cm³

   Volume of cube = l³ = 4 cm × 4 cm × 4 cm = 64 cm³.
4. 3 m

   Volume = L × b × h, so 36 m³ = 6 m × 2 m × h = 12 m² × h. Dividing both sides by 12 m² gives h = 36 m³ ÷ 12 m² = 3 m.

Rating sheet

Mark one score per criterion, then use the notes section for strengths, concerns, and recommendations. The lesson objectives are on the first page for reference.

Revision note ID	cand-bd137435	**Date**	
Subtopic	Three dimensional figures: properties of cylinders and spheres, and volume	**Rater**	

The reference sheet includes the teacher's evaluation questions. A pack that reuses them, reworded or verbatim, may be credited for matching how the lesson was assessed — but reuse on its own does not establish taught-lesson alignment. Rate alignment on the whole of it: the objectives covered, the order taught, what the activities had the students do, the method, and the expected depth.

Criterion (select one score per row)	**1** Poor	**2** Weak	**3** Adequate	**4** Good	**5** Excellent
Objective and curriculum alignment Does the revision note cover the intended lesson objectives and curriculum target?	○	○	○	○	○
Taught-lesson alignment Does it reflect the taught lesson — its sequence, emphasis, activities, and expected depth?	○	○	○	○	○
Scope and conceptual correctness Does it stay within the expected lesson boundary and avoid conceptual/mathematical errors?	○	○	○	○	○
Student-facing clarity and usefulness Can JSS1 learners understand and use it meaningfully as take-home review material?	○	○	○	○	○

Strengths	Concerns	Recommendation

Flag any problems (optional)

Tick all that apply. These are separate from the scores above — leave blank if none apply.

☐	**Misses a required objective** — Skips something the lesson was meant to teach.
☐	**Goes beyond this lesson** — Includes material outside this lesson's scope (e.g. next week's topic).
☐	**Wrong or misleading content** — Contains a mathematical or factual mistake.
☐	**Too hard for JSS1 to follow** — Likely confusing for the students.
☐	**Other:** _____